MOLECULAR GENETICS

Module map

- Module 07: Molecular Genetics
- Connected lab: lab-08_molecular-genetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 07
Molecular Genetics

1. DNA structure
2. Replication
3. Transcription
4. Translation
5. Mutation and gene expression evidence

Lab evidence
lab-08_molecular-genetics.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

MOLECULAR GENETICS

Learning objectives

- Describe DNA structure and base-pairing rules.
- Explain why DNA replication is semiconservative.
- Compare transcription and translation.
- Use codons to connect mRNA sequence to amino acid sequence.
- Explain how mutation can alter protein and trait outcomes.

OBJECTIVE 1
Describe DNA structure and base-pairing rules.

OBJECTIVE 2
Explain why DNA replication is semiconservative.

OBJECTIVE 3
Compare transcription and translation.

OBJECTIVE 4
Use codons to connect mRNA sequence to amino acid sequence.

OBJECTIVE 5
Explain how mutation can alter protein and trait outcomes.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

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Topic sequence

- DNA structure: DNA stores information in nucleotide sequences and complementary base pairing.
- Replication: Replication copies DNA using each strand as a template.
- Transcription: Transcription produces RNA from a DNA gene.
- Translation: Translation uses mRNA codons and ribosomes to build proteins.
- Mutation and gene expression evidence: Mutations and expression patterns connect molecular changes to traits.

01

DNA structure

DNA stores information in nucleotide sequences and complementary base pairing.

02

Replication

Replication copies DNA using each strand as a template.

03

Transcription

Transcription produces RNA from a DNA gene.

04

Translation

Translation uses mRNA codons and ribosomes to build proteins.

05

Mutation and gene expression evidence

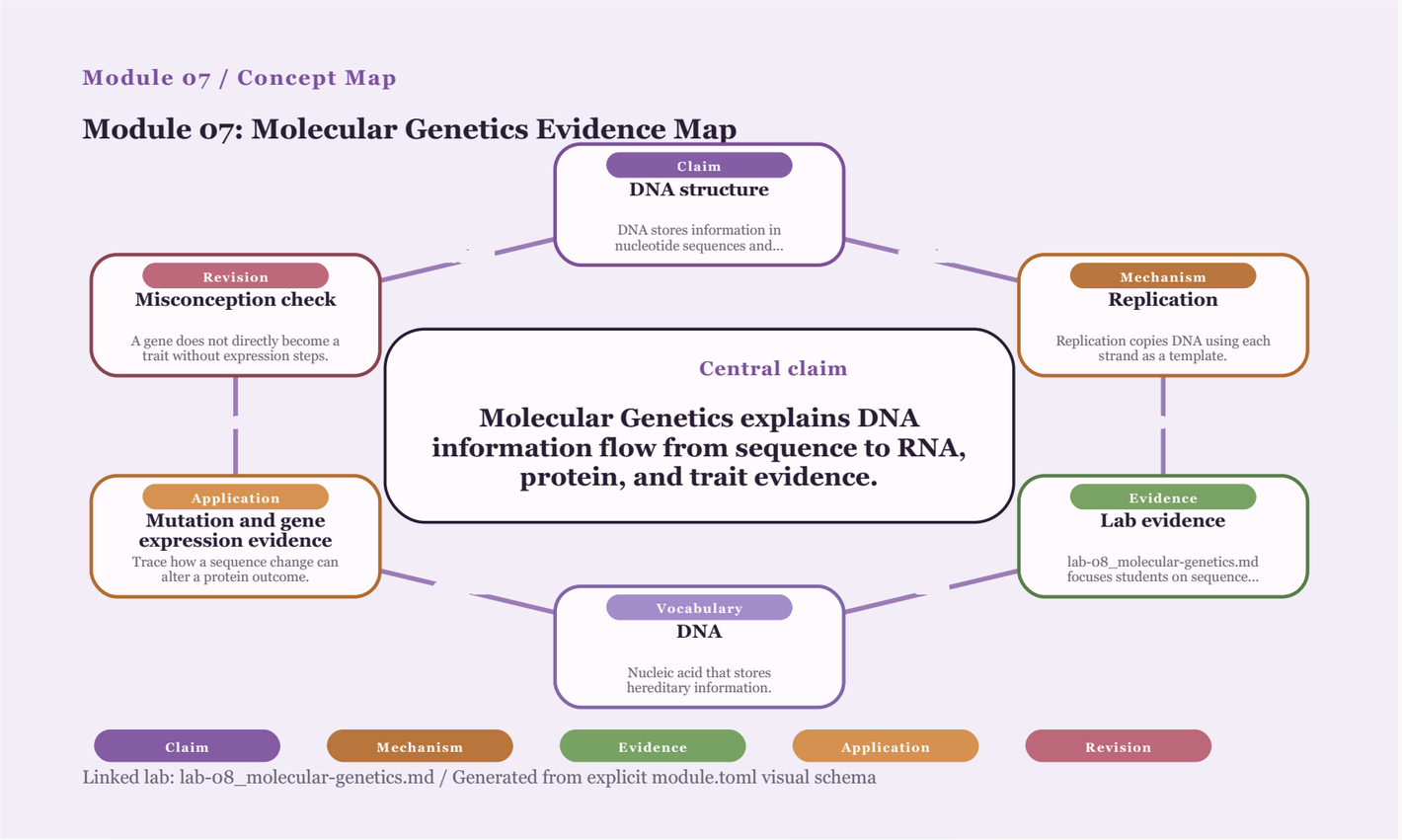
Mutations and expression patterns connect molecular changes to traits.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

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Module 07: Molecular Genetics Evidence Map

- Module focus: Molecular Genetics
- Central claim: Molecular Genetics explains DNA information flow from sequence to RNA, protein, and trait evidence.
- Key nodes to connect: DNA structure, Replication, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-08_molecular-genetics.md supplies an evidence surface for the map.

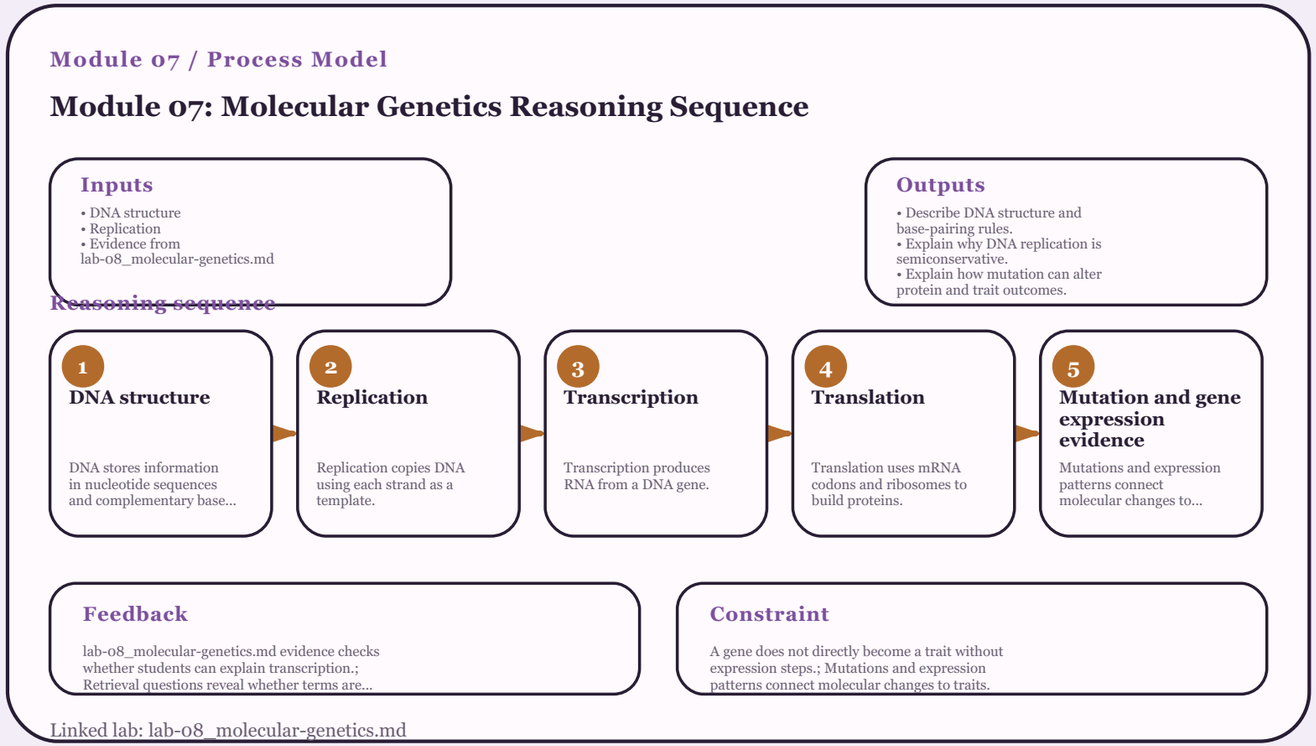


Teaching note: Ask students to explain one edge in their own words before moving on.

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Module 07: Molecular Genetics Reasoning Sequence

- Module focus: Molecular Genetics
- Trace the model: DNA structure -> Replication -> Transcription -> Translation
- Inputs become outputs: DNA structure, Replication -> Describe DNA structure and base-pairing rules., Explain why DNA replication is semiconservative.
- Feedback check: lab-08_molecular-genetics.md evidence checks whether students can explain transcription.
- Lab connection: lab-08_molecular-genetics.md gives students a process to observe or test.



Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

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Terms as evidence handles

- DNA: Nucleic acid that stores hereditary information.
- Nucleotide: DNA or RNA subunit with sugar, phosphate, and base.
- Replication: Process that copies DNA.
- Transcription: Synthesis of RNA from a DNA template.
- Translation: Synthesis of protein from mRNA codons.
- Codon: Three-base mRNA sequence read during translation.

DNA
Nucleic acid that stores hereditary information.

NUCLEOTIDE
DNA or RNA subunit with sugar, phosphate, and base.

REPLICATION
Process that copies DNA.

TRANSCRIPTION
Synthesis of RNA from a DNA template.

TRANSLATION
Synthesis of protein from mRNA codons.

CODON
Three-base mRNA sequence read during translation.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

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Lab connection

- Lab file: lab-08_molecular-genetics.md
- Lab evidence should test or illustrate: Transcription
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
DNA structure

EVIDENCE
lab-08_molecular-genetics.md

CLAIM
Describe DNA structure and base-pairing rules.

Teaching note: Make the lab connection explicit before students encounter the activity.

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Contrast check

- Do not stop at: DNA stores information in nucleotide sequences and complementary base pairing.
- Stronger explanation: Mutations and expression patterns connect molecular changes to traits.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

DNA stores information in nucleotide sequences and complementary base pairing.

DEEPER

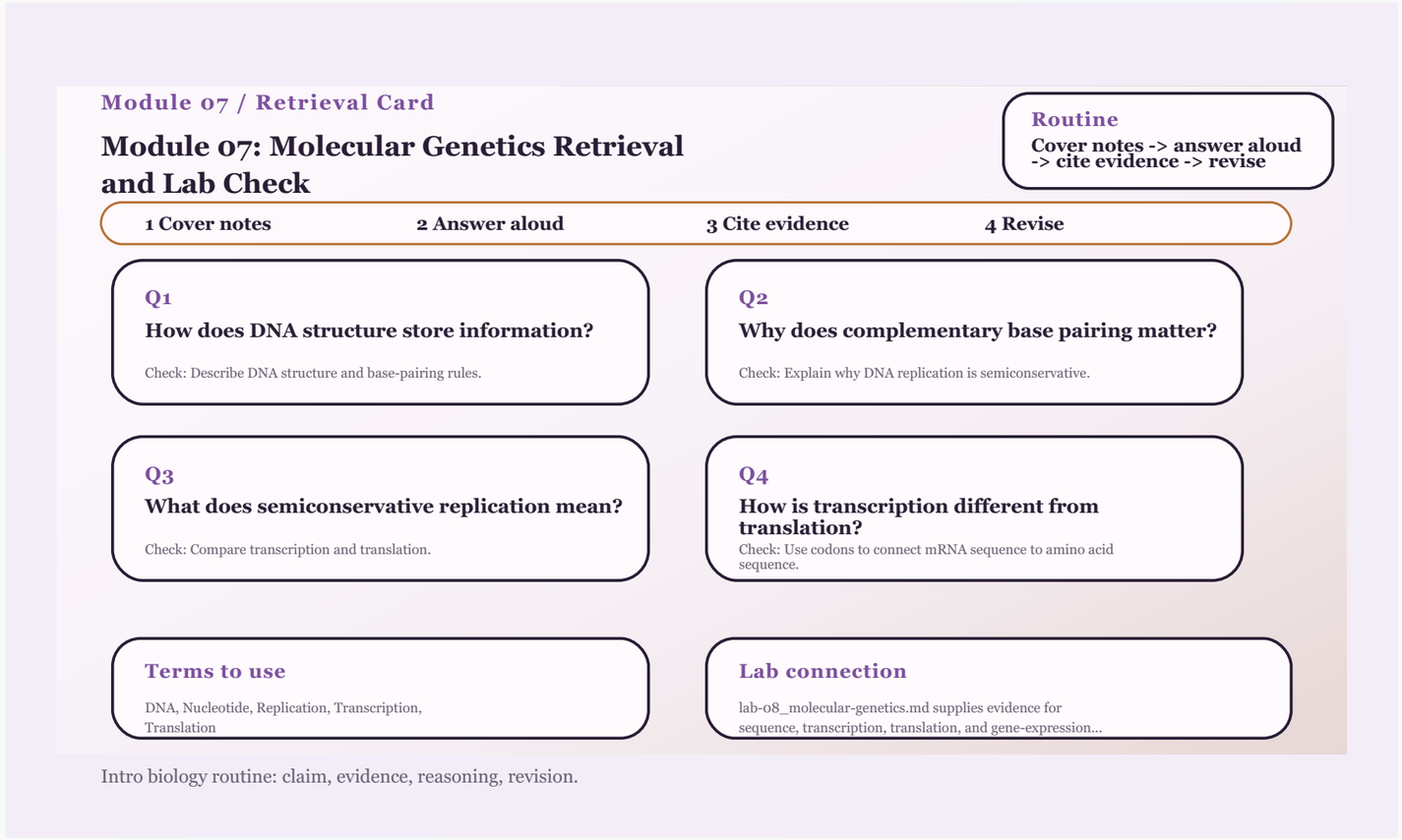
Mutations and expression patterns connect molecular changes to traits.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

MOLECULAR GENETICS

Module 07: Molecular Genetics Retrieval and Lab Check

- Module focus: Molecular Genetics
- Retrieval prompt: How does DNA structure store information?
- Answer check: Describe DNA structure and base-pairing rules.
- Required terms: DNA, Nucleotide, Replication
- Lab connection: lab-08_molecular-genetics.md; lab-08_molecular-genetics.md supplies evidence for sequence, transcription, translation, and gene-expression evidence.



Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

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Practice quiz bridge

- 1. In DNA, the base adenine always pairs with:
- 2. DNA replication is described as "semiconservative" because each new double helix contains:
- 3. Which sequence shows the normal flow of genetic information?
- 4. A mutation changes a single DNA base. Why might this matter?

QUESTION 1

Key: C

In DNA, the base adenine always pairs with:

QUESTION 2

Key: A

DNA replication is described as "semiconservative" because each new double helix contains:

QUESTION 3

Key: D

Which sequence shows the normal flow of genetic information?

QUESTION 4

Key: B

A mutation changes a single DNA base. Why might this matter?

Teaching note: Use the quiz as formative feedback. Answer key: 1=C, 2=A, 3=D, 4=B

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Synthesis and exit ticket

- Exit claim: explain dna structure using evidence from lab-08_molecular-genetics.md.
- Must include: DNA, Nucleotide, Replication.
- Revision prompt: what evidence would change your explanation?

CLAIM
DNA structure

EVIDENCE
lab-08_molecular-genetics.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.